Fullstack Personal Expense & Finance Tracker

System Architecture Specification & Technical Documentation — Category: Full Stack Financial Analytics

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Repository: https://github.com/rakshapatil526/Expense-Tracker-

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1. Executive Summary & Financial Architecture

The Fullstack Expense Tracker is a personal finance management platform engineered with React.js, Django REST Framework, and PostgreSQL. It delivers financial health visibility through categorized income and expense logging, automated monthly budget comparisons, and interactive Recharts data visualizations.

Containerized via Docker and Docker Compose, the application demonstrates production container orchestration.

2. Database Schema & Data Models

Table	Primary Key	Schema Columns & Foreign Keys
users	user_id (BigAutoField)	name, email (unique, indexed), password (PBKDF2 SHA-256), created_at
categories	category_id (BigAuto)	user_id (FK: users, nullable for system defaults), category_name, type (income/expense), icon, color
transactions	transaction_id (BigAuto)	user_id (FK: users), category_id (FK: categories), title, amount (decimal), transaction_date, notes

3. Analytics & Visualization Modules

• **Financial Health KPIs:** Net balance, total monthly income, total monthly expenditure, and savings rate percentage.

• **Spending Breakdown Donut:** Category-wise percentage distribution rendered dynamically using Recharts.

• **Monthly Cash Flow Trend:** Interactive area chart displaying daily income versus expense trajectory.

• **Data Export:** Filterable transaction history with CSV spreadsheet export capability.

4. Containerization & Tech Stack

• **Stack:** React.js (Vite), Python 3.13, Django 5.2, DRF, SimpleJWT, Recharts, Lucide Icons, Docker Compose.

• **Verification:** Verified directly from repository structure and database models.